

Name: _____

Class: file

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

2007

EXTENSION 1 MATHEMATICS

Instructions:

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start each question on a new page

Total Marks – 84

- Attempt Questions 1-7
- All questions are of equal value

(For markers use only)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total

Question 1 (12 marks)

- a) Find $\log_2 3$ correct to 3 decimal places. 1
- b) i) Sketch $y = |2x|$ 1
- ii) By drawing suitable lines on your sketch above, determine that one of the following equations $A: |2x| = x - 1$ and $B: |2x| = 1 - x$ has no solutions and solve the other. 3
- c) Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{\frac{1}{2}x}$ 1
- d) If α, β and δ are the roots of $2x^3 + 12x^2 - 6x + 1 = 0$ find the values of
- i) $\alpha + \beta + \delta$ 1
- ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\delta}$ 2
- e) Use the substitution $u = 4 - x^2$ or otherwise to find $\int x\sqrt{4 - x^2} dx$ 3

Question 2 (12 marks) (start a new page)

- a) Given that $\log_x 2 = a$ and $\log_x 3 = b$ find $\log_x 2.25$ in terms of a and b . 2
- b) Evaluate $\int_{-1}^2 |1 - 2x| dx$ by considering a graph or otherwise. 2
- c) Find i) $\int \frac{3x}{x^2 + 1} dx$ 2
- ii) $\int \frac{3}{x^2 + 1} dx$ 2
- d) Solve $\sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$ 2
- e) An area of 1 unit² is bounded by the curve $y = \frac{1}{x}$, the x axis and the lines $x = e$ and $x = k$ 2
- Find the value of k (in exact form), if $k > e$.

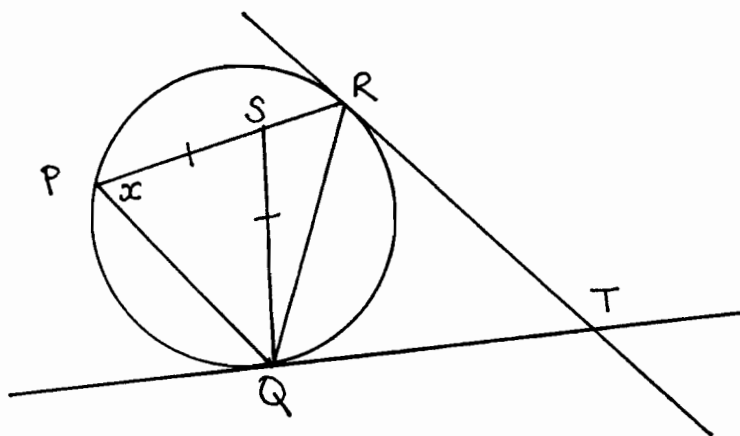
Question 3 (12 marks) (start a new page)

a) Find $\int \cos^2 3x \, dx$ 2

b) i) Show that $\tan 75^\circ = \sqrt{3} + 2$ 2

ii) The lines $y = mx$ and $x = y\sqrt{3}$ meet at an angle of 75° . Find only one value of m . 2

c) PQR is a triangle inscribed in a circle. S is a point on PR , chosen so that $QS = SP$. Tangents from an external point T touch the circle at Q and R . Copy the diagram onto your page and prove that the quadrilateral $QTRS$ is cyclic. Let $\angle SPQ = x$



d) i) Show that $\frac{d}{dx} [\tan^{-1}(e^x) + \tan^{-1}(e^{-x})] = 0$ 2

ii) Hence evaluate $\tan^{-1}(e^x) + \tan^{-1}(e^{-x})$ for all values of x . 1

Question 4 (12 marks) (start a new page)

a) If $y = xe^x$

i) Prove $\frac{dy}{dx} = e^x(x+1)$ and $\frac{d^2y}{dx^2} = e^x(x+2)$ 2

ii) Hence prove by mathematical induction for all positive integers n , that

$$\frac{d^n y}{dx^n} = e^x(x+n) \quad 3$$

b) For the curve $y = 2\sin^{-1}(1-4x)$, state the domain and range and sketch the graph. 3

c) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$. The line ℓ is a tangent at P

i) Write the equation of ℓ 1

ii) If ℓ meets the y axis at A , show that $SP=SA$ where S is the focus of the parabola. 2

iii) Hence show that ℓ is equally inclined to SP and the axis of the parabola. 1

Question 5 (12 marks) (start a new page)

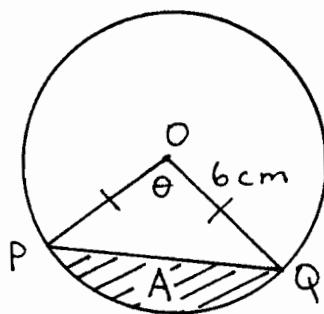
a) i) The polynomial equation $P(x) = 0$ has a double root at $x = a$. By putting

$$P(x) = (x-a)^2 \cdot Q(x) \text{ show that } P'(a) = 0 \quad 2$$

ii) You are told the polynomial $P(x) = mx^4 + nx^3 - 6x^2 + 22x - 12$ has a double root at $x = 1$. Find the value of m and n . 3

b) O is the centre of a circle with radius 6cm.

$$\angle POQ = \theta \text{ radians}$$



i) Find an expression for A , the area of the minor segment, cut off by the chord PQ , in terms of θ . 1

ii) If θ is increasing at 0.75 radians/second, what is the rate of change of A when

$$\theta = \frac{\pi}{3} ? \quad 2$$

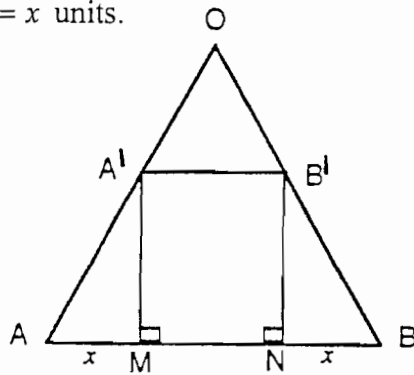
- c) Katrina, a sky-diver, opens her parachute when falling at 30m/s. Thereafter her acceleration is given by $\frac{dv}{dt} = k(6 - v)$ where k is a constant.
- Show that this condition is satisfied when $v = 6 + Ae^{-kt}$ and find the constant A . 2
 - One second after opening her chute, her velocity has fallen to 10.7 m/s. Find the value of k correct to 2 decimal places. 1
 - Find her velocity, correct to 1 decimal place, two seconds after her chute has opened. 1

Question 6 (12 marks) (start a new page)

- Show that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \frac{d^2x}{dt^2}$ 1
 - An object is falling through a fluid in such a way that its acceleration is given by, $\frac{d^2x}{dt^2} = \frac{4}{\sqrt{x}}$ where x is the distance the object has fallen in metres and t is time in seconds. 3.
If the object started from rest, how fast would it be travelling after falling through a distance of 7 metres. (to 1 decimal place)?
- Sketch the function $f(x) = x + \frac{1}{x}$ for $x > 0$ showing the stationary point and asymptotes. 2
 - State the largest possible domain containing $x = 2$ for which $f(x)$ has an inverse $f^{-1}(x)$. 1
 - Sketch $y = f^{-1}(x)$ on the diagram above. 1
 - Show that $f^{-1}(x) = \frac{x}{2} + \frac{1}{2}\sqrt{x^2 - 4}$ 2
 - Assume $x = N$, when N is not in the domain chosen for part ii) but still in the domain for $f(x)$.
Find $f^{-1}[f(N)]$ 2

Question 7 (12 marks) (start a new page)

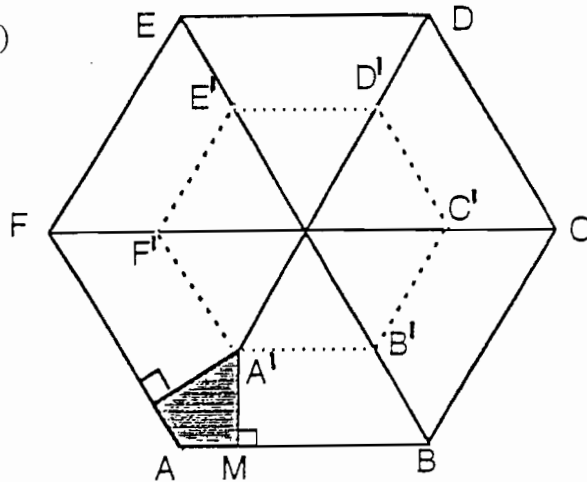
- a) i) OAB is an equilateral triangle side m units. $A^1B^1 \parallel AB$ and $AM = NB = x$ units.



Show that the area of $\triangle OA^1B^1$ is given by $\frac{\sqrt{3}(m-2x)^2}{4}$

2

ii)



Use part i) as
you answer part ii)

$ABCDEF$ is a regular hexagon, side m units. The sides of $A^1B^1C^1D^1E^1F^1$ are parallel to those of $ABCDEF$. From each vertex, portions such as the one shaded are removed. The remainder is folded along the dotted lines to form a hexagonal prism

α) If $AM = x$ units prove the volume of the prism is given by

$$V = \frac{9x(m-2x)^2}{2} \text{ units}^3 \quad 2$$

β) Prove that the maximum volume of such a prism is $\frac{m^3}{3} \text{ units}^3$ 3

- b) If $\tan 2x = \frac{\tan x}{a \tan x + b}$ and $\tan x \neq 0$

i) Find a condition in terms of a and b , for the equation above, to have two different roots $\tan \alpha$ and $\tan \beta$

2

ii) Assuming this condition to be satisfied prove $\tan^2(\alpha - \beta) = \frac{a^2 - 2b + 1}{a^2 + 2b + 1}$

3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

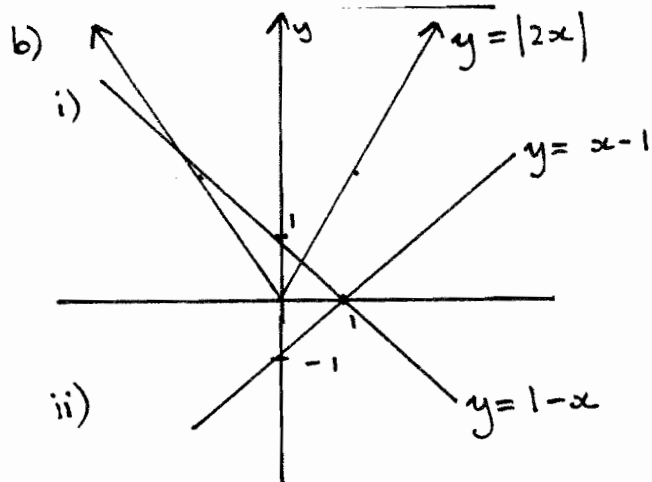
NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

a)

$$\log_2 3 = \frac{\log_e 3}{\log_e 2}$$

$$= \underline{\underline{1.585 \text{ (3 dec. pl.)}}}$$



$\therefore |2x| = 1 - x$ no solutions; no pts of intersection

$|2x| = 1 - x$ 2 solutions

$$2x = 1 - x \quad 2x = -(1 - x)$$

$$x = \frac{1}{3} \quad \text{and} \quad x = -1$$

c)

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{\frac{1}{2}x}$$

$$= \lim_{x \rightarrow 0} \sin 2x \cdot \underline{\underline{2}}$$

d) $a=2 \quad b=12 \quad c=-6 \quad d=1$

i) $\alpha + \beta + \gamma = \frac{-b}{a} = \frac{-12}{2} = \underline{\underline{-6}}$

ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$$= \frac{\frac{c}{a}}{\frac{-d}{a}} = \frac{-3}{-1/2}$$

$$= \underline{\underline{6}}$$

e) $u = 4 - x^2$

$$\frac{du}{dx} = -2x$$

$$du = -2x \, dx$$

$$\therefore dx = \frac{du}{-2x}$$

$$\int x \sqrt{4 - x^2} \, dx = \int x \sqrt{u} \cdot \frac{du}{-2x}$$

$$= -\frac{1}{2} \int u^{1/2} \, du$$

$$= -\frac{1}{2} \left[\frac{2u^{3/2}}{3} \right] + C$$

$$= \underline{\underline{-\frac{1}{3} \sqrt{(4 - x^2)^3} + C}}$$

Question 2

a) $\log_x 2.25 = \log_x \frac{9}{4}$

$$= \log_x 9 - \log_x 4$$

$$= 2 \log_x 3 - 2 \log_x 2$$

$$= \underline{\underline{2b - 2a}}$$

b)

i)

ii)

$$\therefore \int^2 |1 - 2x| \, dx$$

(2)

$$c) i) \int \frac{3x}{x^2+1} dx = \frac{3}{2} \int \frac{2x}{x^2+1} dx$$

$$= \frac{3}{2} \ln(x^2+1) + c$$

$$ii) \int \frac{3}{x^2+1} dx = 3 \int \frac{1}{x^2+1} dx$$

$$= 3 \tan^{-1} x + c$$

$$d) \sin 2\theta = \sin \theta$$

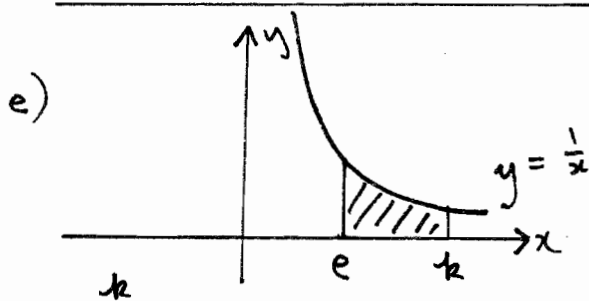
$$2 \sin \theta \cdot \cos \theta = \sin \theta$$

$$2 \sin \theta \cdot \cos \theta - \sin \theta = 0$$

$$\sin \theta (2 \cos \theta - 1) = 0$$

$$\sin \theta = 0 \quad \cos \theta = \frac{1}{2}$$

$$\theta = 0, \pi, 2\pi \quad \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$



$$\int_e^k \frac{1}{x} dx = 1$$

$$\left[\ln x \right]_e^k = 1$$

$$\ln k - \ln e = 1$$

$$\ln k - 1 = 1$$

$$\log_e k = 2$$

$$k = e^2$$

$$\therefore \int \cos^2 3x dx = \frac{1}{2} \int (\cos 6x + 1) dx$$

$$= \frac{1}{2} \left[\frac{1}{6} \sin 6x + x \right] + c$$

$$= \frac{1}{12} \sin 6x + \frac{x}{2} + c$$

$$b) i) \tan 75^\circ = \tan (30^\circ + 45^\circ)$$

$$= \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \cdot \tan 45^\circ}$$

$$= \left(\frac{1}{\sqrt{3}} + 1 \right) \div \left(1 - \frac{1}{\sqrt{3}} \right)$$

$$= \frac{1 + \sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3} - 1}$$

$$= \frac{1 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{2\sqrt{3} + 4}{2}$$

$$= \sqrt{3} + 2$$

$$ii) \text{gradients are } m \text{ and } \frac{1}{\sqrt{3}}$$

$$\tan 75 = \frac{m - \frac{1}{\sqrt{3}}}{1 + \frac{m}{\sqrt{3}}} \quad \left(\begin{array}{l} \text{take the} \\ \text{case only} \\ \text{one solution} \\ \text{required} \end{array} \right)$$

$$\sqrt{3} + 2 = \frac{m - \frac{1}{\sqrt{3}}}{1 + \frac{m}{\sqrt{3}}}$$

$$(\sqrt{3} + 2) \left(1 + \frac{m}{\sqrt{3}} \right) = m - \frac{1}{\sqrt{3}}$$

$$\sqrt{3} + m + 2 + \frac{2m}{\sqrt{3}} = m - \frac{1}{\sqrt{3}}$$

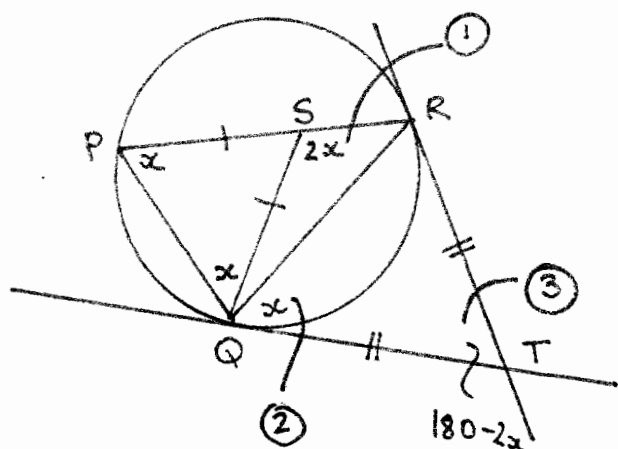
$$\frac{2m}{\sqrt{3}} = -\frac{1}{\sqrt{3}} - \sqrt{3} - 2$$

$$2m = -1 - 3 - 2\sqrt{3}$$

$$m = -2 - \sqrt{3}$$

Question 3

c)



$$\hat{RSQ} = 2x \text{ (ext. angle of isosceles triangle)}$$

$$\hat{RQT} = x \text{ (alt. segment theorem)}$$

since $QT = TR$ (tangents from an external pt are =)

$$\hat{RTQ} = 180 - 2x \text{ (angle sum of isosceles triangle)}$$

$$\hat{RSQ} + \hat{RTQ} = 180^\circ$$

$\therefore$ QTRS is cyclic since opposite angles are supp.

d) i) $\frac{d}{dx} (\tan^{-1} e^x + \tan^{-1} e^{-x})$

$$= \frac{e^{2x}}{e^{2x} + 1} - \frac{e^{-x}}{e^{-2x} + 1}$$

$$= \frac{e^x}{e^{2x} + 1} - \left[\frac{1}{e^x} \div \left(\frac{1}{e^{2x}} + 1 \right) \right]$$

$$= \frac{e^x}{e^{2x} + 1} - \left[\frac{1}{e^x} \times \frac{e^{2x}}{1 + e^{2x}} \right]$$

$$= e^{2\alpha} - e^{\alpha}$$

ii) subst. $x=0$ since true for all x
 $\tan^{-1}(e^0) + \tan^{-1}(e^0)$

$$2 \tan^{-1} 1 = 2 \times \frac{\pi}{4}$$

$$= \pi/2$$

Question 4

a) $y = x e^x$ $u = x \quad v = e^x$
 i) $\frac{dy}{dx} = e^x + x e^x$ $u' = 1 \quad v' = e^x$

$$\frac{dy}{dx} = e^x (1+x)$$

$$\frac{d^2y}{dx^2} = e^x + e^x(1+x)$$

$$= e^x (1 + 1 + x)$$

$$\therefore \frac{d^2y}{dx^2} = e^x(2+x)$$

ii) Step 1: Show true for $n=1$

$$\therefore \frac{dy}{dx} = e^x(x+1) \text{ from above}$$

Step 2 : Assume true for $n=k$
some +ve integer

$$\frac{d^k y}{dx^k} = e^x (x + k)$$

Step 3: Prove true for $n = k+1$

ie show that $\frac{d y^{(k+1)}}{d x^{(k+1)}} = e^{x^{(k+1)}} (x + k+1)$

$$\text{LHS} = \frac{d^{(k+1)}y}{dx^{(k+1)}}$$

$$= \frac{d}{dx} (e^x (x+k)) \text{ from Step 2}$$

$$= \frac{d}{dx} (x \cdot e^x + k e^x)$$

$$= e^x + x e^x + k e^x$$

$$= e^x (1 + x + k)$$

$$= \text{RHS}$$

Step 4: Since true for $n=1$
and if assumed true for $n=k$
(some +ve integer) we have
shown true for $n=k+1$
 $\therefore$ true for all +ve integers

$$b) \quad y = 2 \sin^{-1}(1-4x)$$

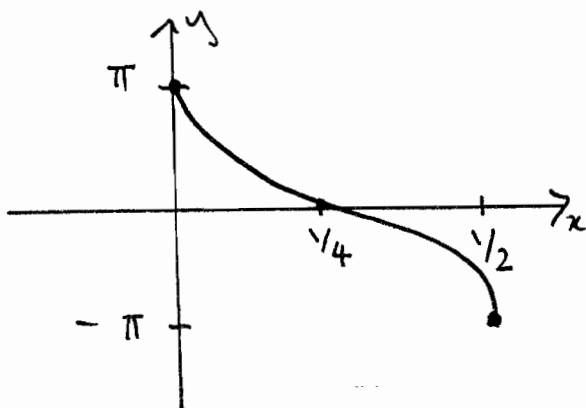
$$\frac{y}{2} = \sin^{-1}(1-4x)$$

$$-\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$$

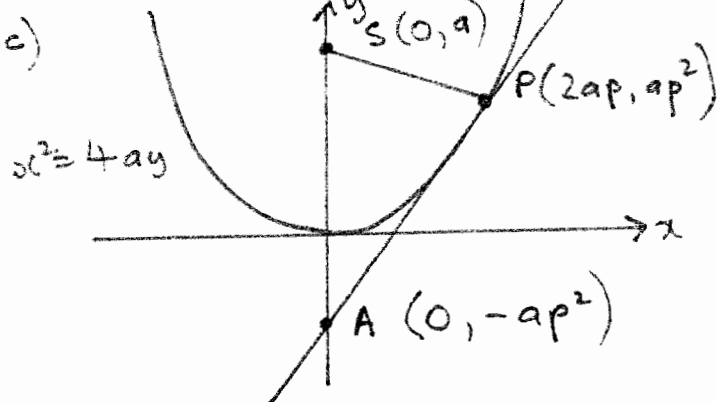
$$\therefore \text{Range: } -\pi \leq y \leq \pi$$

$$-1 \leq 1-4x \leq 1$$

$$\text{Domain: } 0 \leq x \leq \frac{1}{2}$$



c)



$$i) \quad y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

$$m_P = \frac{2ap}{2a} = p$$

$$\text{tang: } y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap$$

$$\underline{\underline{y = px - ap^2}}$$

$$ii) \quad A(0, -ap^2)$$

$$SA = a + ap^2 = \underline{\underline{a(1+p^2)}}$$

$$SP = \sqrt{(2ap-0)^2 + (ap^2-a)^2}$$

$$= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$$

$$= a\sqrt{p^4 + 2p^2 + 1}$$

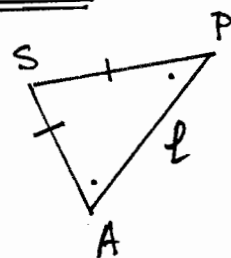
$$= a\sqrt{(p^2+1)^2}$$

$$\underline{\underline{SP = a(p^2+1)}}$$

$$iii) \quad \therefore SP = SA$$

$\triangle SPA$ is
isosceles

$$\therefore \hat{SAP} = \hat{SPA}$$



Question 5

a) i) $P(x) = (x-a)^2 \cdot Q(x)$

$$u = (x-a)^2 \quad v = Q(x)$$

$$u' = 2(x-a) \quad v' = Q'(x)$$

$$P'(x) = 2(x-a) \cdot Q(x) + (x-a)^2 \cdot Q'(x)$$

$$P'(a) = 2(a-a)Q(a) + (a-a)^2 \cdot Q'(a)$$

$$\therefore P'(a) = 0$$

ii) $P(1) = 0$ and $P'(1) = 0$

$$P(x) = mx^4 + nx^3 - 6x^2 + 22x - 12$$

$$P(1) = m + n - 6 + 22 - 12 = 0$$

$$m + n = -4 \quad \text{--- (1)}$$

$$P'(x) = 4mx^3 + 3nx^2 - 12x + 22$$

$$P'(1) = 4m + 3n - 12 + 22 = 0$$

$$4m + 3n = -10 \quad \text{--- (2)}$$

① $\times 4$: $4m + 4n = -16$

② $\underline{4m + 3n = -10}$

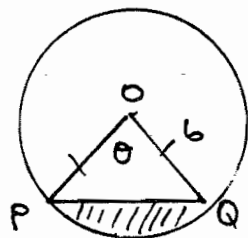
$$n = -6$$

$$m - 6 = -4$$

$$m = 2$$

$$\therefore \underline{\underline{m = 2, n = -6}}$$

b)



i) $A = \frac{1}{2} \cdot 6^2 (\theta - \sin \theta)$

$$\underline{\underline{A = 18\theta - 18 \sin \theta}}$$

ii) $\frac{d\theta}{dt} = .75$ require $\frac{dA}{dt}$
when $\theta = \pi/2$

$$\frac{dA}{d\theta} = 18 - 18 \cos \theta$$

$$\therefore \frac{dA}{dt} = .75 (18 - 18 \cos \frac{\pi}{3})$$

$$= .75 (18 - 18 \cdot \frac{1}{2})$$

$$\frac{dA}{dt} = \underline{\underline{6.75 \text{ cm}^2/\text{second}}}$$

c) i) $V = 6 + Ae^{-kt}$

$$\therefore \frac{dV}{dt} = -k(Ae^{-kt})$$

$$= -k(V - 6)$$

$$= -kV + 6k$$

$$\therefore \frac{dV}{dt} = k(6 - V)$$

as required

$$V = 30 \text{ m/s } t = 0 \text{ sub.}$$

$$\therefore 30 = 6 + Ae^0$$

$$\therefore \underline{\underline{A = 24}}$$

ii) $t = 1 \quad V = 10.7$

$$V = 6 + 24e^{-kt}$$

$$10.7 = 6 + 24e^{-k}$$

$$4.7 = 24e^{-k}$$

$$\ln \left(\frac{4.7}{24} \right) = -k$$

$$\underline{\underline{k = 1.63}} \quad (2 \text{ dec. pl})$$

iii) $t = 2$

$$V = 6 + 24e^{-1.63 \times 2}$$

$$\underline{\underline{V = 6.9 \text{ m/s}}} \quad (1 \text{ dec. pl})$$

6

Question 6

$$\begin{aligned}
 \text{a) i) } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \cdot \frac{dv}{dx} \\
 &= v \cdot \frac{dv}{dx} \\
 &= \frac{dx}{dt} \cdot \frac{dv}{dx} \\
 &= \frac{dv}{dt} \\
 &= \ddot{x}
 \end{aligned}$$

$$\text{ii) } \ddot{x} = \frac{4}{\sqrt{x}}$$

$$t=0, \dot{x}=0, x=0$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4x^{-1/2}$$

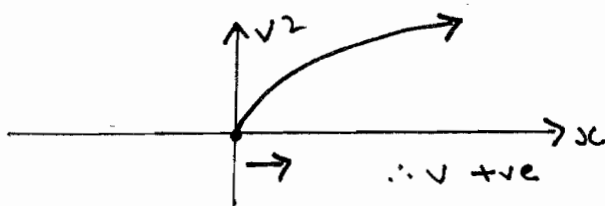
$$\frac{1}{2} v^2 = \frac{4x^{1/2}}{1/2} + c_1$$

$$= 8\sqrt{x} + c_1$$

$$v^2 = 16\sqrt{x} + c$$

$$\text{sub } x=0, v=0 \therefore c=0$$

$$v^2 = 16\sqrt{x}$$



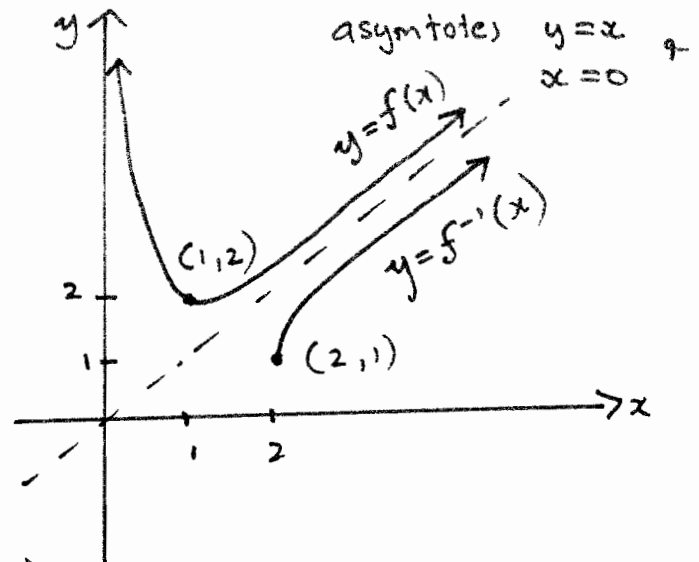
$$\text{sub. } x=7$$

$$v^2 = 16\sqrt{7}$$

$$v = 6.5 \text{ m/s}$$

$$\begin{aligned}
 \text{b) i) } f(x) &= x + x^{-1} \\
 f'(x) &= 1 - x^{-2}
 \end{aligned}$$

since \$x > 0\$ at \$(1,2)\$ \$f''(x) > 0\$ min



$$\text{ii) } \underline{\underline{D: x \geq 1}}$$

iii) see diagram

$$\text{iv) } x = y + \frac{1}{y}$$

$$xy = y^2 + 1$$

$$-1 = y^2 - xy$$

$$-4 = 4y^2 - 4xy$$

$$-4 + x^2 = 4y^2 - 4xy + x^2$$

$$-4 + x^2 = (2y - x)^2$$

$$\pm \sqrt{x^2 - 4} = 2y - x$$

from domain above take \$+\sqrt\$

$$2y = x + \sqrt{x^2 - 4}$$

$$\therefore y = \frac{x}{2} + \frac{1}{2} \sqrt{x^2 - 4}$$

v) $f^{-1}[f(N)]$

N not in above domain

$\therefore$ use $f^{-1}(x) = \frac{x}{2} - \frac{1}{2}\sqrt{x^2-4}$

$f(N) = N + \frac{1}{N}$

$f^{-1}(f(N)) = \frac{1}{2}\left(N + \frac{1}{N}\right) - \frac{1}{2}\sqrt{\left(N + \frac{1}{N}\right)^2 - 4}$

$= \frac{N}{2} + \frac{1}{2N} - \frac{1}{2}\sqrt{N^2 + 2 + \frac{1}{N^2} - 4}$

$= \frac{N}{2} + \frac{1}{2N} - \frac{1}{2}\sqrt{N^2 - 2 + \frac{1}{N^2}}$

$= \frac{N}{2} + \frac{1}{2N} - \frac{1}{2}\sqrt{\left(N - \frac{1}{N}\right)^2}$

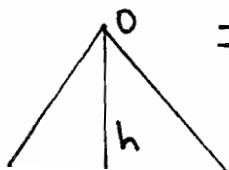
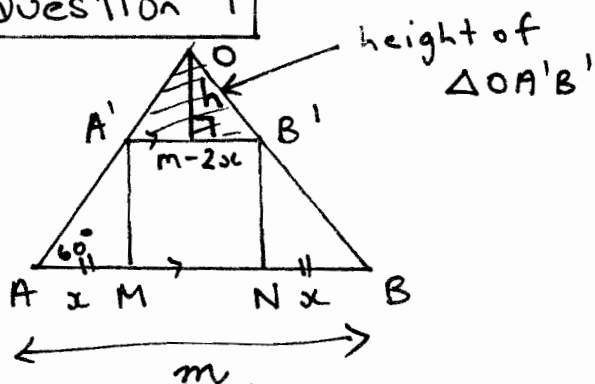
N.B.
maintain
the
-ve here

$= \frac{N}{2} + \frac{1}{2N} - \frac{1}{2}\left(N - \frac{1}{N}\right)$

$= \cancel{\frac{N}{2}} + \frac{1}{2N} - \cancel{\frac{N}{2}} + \frac{1}{2N}$

$f^{-1}(f(N)) = \frac{1}{N}$

Question 7



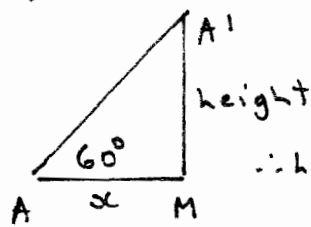
In $\Delta OA'B'$
 $\tan 60^\circ = \frac{h}{\frac{m-2x}{2}}$

$\frac{\sqrt{3}(m-2x)}{2} = h$

$\therefore$ Area of $OA'B' = \frac{1}{2}(m-2x)\frac{\sqrt{3}(m-2x)}{2}$

$A = \frac{\sqrt{3}}{4}(m-2x)^2$

ii) $V = \text{Abase} \times \text{height}$



$\therefore \text{height} = x \tan 60^\circ = x\sqrt{3}$

$V = 6\left(\frac{\sqrt{3}}{4}(m-2x)^2\right) \cdot x\sqrt{3}$

$V = \frac{9}{2}x(m-2x)^2$

iii) $\hookrightarrow u = \frac{9x}{2} \quad v = (m-2x)^2$
 $u' = \frac{9}{2} \quad v' = 2 \cdot -2(m-2x)$
 $v' = -4(m-2x)$

using product rule

$V' = \frac{9}{2}(m-2x)^2 - 18x(m-2x)$

$V' = 9(m-2x)\left[\frac{1}{2}(m-2x) - 2x\right]$

$V' = 9(m-2x)\left(\frac{m}{2} - 3x\right)$

st pt $V' = 0$

$\therefore x = \frac{m}{2} \text{ OR } x = \frac{m}{6}$

test max/min

x	m/8	m/6	m/4	m/2	m
V'	+	0	-	0	+
	+ / max		- / min		+

β) sub $\alpha = \frac{m}{6}$ into V

$$V = \frac{9}{2} \cdot \frac{m}{6} \cdot \left(m - 2\frac{m}{6}\right)^2$$

$$= \frac{3m}{4} \left(\frac{2m}{3}\right)^2$$

$$= \frac{3m}{4} \cdot \frac{4m^2}{9}$$

$$\text{max. } V = \frac{m^3}{3} \text{ unit}^3$$

b) i) different roots if $\Delta > 0$

$$\tan 2x = \frac{\tan x}{a \tan x + b}$$

$$\frac{2 \tan x}{1 - \tan^2 x} = \frac{\tan x}{a \tan x + b}$$

$$2 \tan x (a \tan x + b) = \tan x (1 - \tan^2 x)$$

since $\tan x \neq 0$ given

$$2(a \tan x + b) = 1 - \tan^2 x$$

$$2a \tan x + 2b = 1 - \tan^2 x$$

$$\tan^2 x + 2a \tan x + 2b - 1 = 0$$

$$\Delta = (2a)^2 - 4 \cdot 1 \cdot (2b - 1)$$

require $\Delta > 0$

$$4a^2 - 8b + 4 > 0$$

$$\underline{\underline{a^2 - 2b + 1 > 0}}$$

$$\begin{aligned} \text{ii) LHS} &= \tan^2(\alpha - \beta) \\ &= [\tan(\alpha - \beta)]^2 \end{aligned}$$

$$= \left(\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta} \right)^2$$

$$= \frac{\tan^2 \alpha + \tan^2 \beta - 2 \tan \alpha \cdot \tan \beta}{(1 + \tan \alpha \cdot \tan \beta)^2}$$

$$\left(\text{since } A^2 + B^2 = (A+B)^2 - 2AB \right)$$

$$\Rightarrow = \frac{(\tan \alpha + \tan \beta)^2 - 4 \tan \alpha \cdot \tan \beta}{(1 + \tan \alpha \cdot \tan \beta)^2}$$

$$= \frac{(\text{sum of roots})^2 - 4 \times \text{product of roots}}{(1 + \text{product of roots})^2}$$

$$= \frac{(-2a)^2 - 4(2b-1)}{(1 + (2b-1))^2}$$

$$= \frac{4a^2 - 8b + 4}{4b^2}$$

$$= \frac{a^2 - 2b + 1}{b^2}$$